

Kernel Module Configuration

Objectives

After completing this lesson, you should be able to:

- Describe loadable kernel modules
- Dynamically load and unload kernel modules
- Configure kernel module parameters



Loadable Kernel Modules (LKM)

- LKM extend the functionality of a running kernel.
- Kernel modules are dynamically loaded and unloaded.
- To list currently loaded kernel modules:

```
# lsmod
```

- To view details about a specific kernel module:

```
# modinfo <module_name>
```

- In UEK R4, kernel module file names have a ".ko" extension, for "kernel object".
 - For example, the `ext4` kernel module name in UEK R4 is `ext4.ko`.
- In UEK R5, kernel modules are compressed, and have ".xz" added to ".ko".
 - The `ext4` kernel module name in UEK R5 is `ext4.ko.xz`.
- Kernel modules often have dependencies.

Loading and Unloading Kernel Modules

- To load kernel modules:

```
# modprobe <module_name>
```

- To unload kernel modules:

```
# modprobe -r <module_name>
```

- Kernel module dependencies are listed in `/lib/modules/<kernel_version>/modules.dep`.
 - The file is created by `depmod` when kernel modules are installed.
- The Oracle-supplied kernel modules used in Grid Infrastructure are:
 - `oracleacfs`, `oracleadvn`, `oracleoks`
- The directory to specify modules to load at boot time is:
 - `/etc/sysconfig/modules`

Kernel Module Parameters

- Pass parameters to a kernel module:

```
# modprobe <module_name> [parameter=value]
```

- Configuration directory for modprobe:
 - /etc/modprobe.d
- Create *.conf files in /etc/modprobe.d to:
 - Specify options
 - Create aliases
 - Override normal modprobe behavior
 - Blacklist kernel modules
- Valid commands to use in these files include:
 - alias, options, install, remove, blacklist

Quiz

Which of the following commands displays details about a kernel module?

- a. `lsmod <module_name>`
- b. `modinfo <module_name>`
- c. `modprobe <module_name>`
- d. `depmod <module_name>`



Summary

In this lesson, you should have learned how to:

- Describe loadable kernel modules
- Dynamically load and unload kernel modules
- Configure kernel module parameters



Practice 9: Overview

This practice covers using loadable kernel modules.